

AP CSP | Lesson 06: Compression

Homework | Multiple-choice and written practice

Name: _____ Class: _____ Date: _____

Time: 40 minutes | Total: 30 teacher marks (not an AP score) | Show working.

Original AP-style MCQs and classroom written practice; these are not official AP questions.

Complete all 12 original questions (30 marks, about 40 minutes). Q5 requires exactly two choices. For toy RLE, each original uppercase letter uses one byte and each count-letter pair uses two bytes; counts are 1-9. Ignore headers unless stated. Show the original size, compressed size and the original-size denominator when calculating a percentage saved.

1. A record occupies 18 bytes before compression. How many bits is this? [1 mark]

A. 26 B. 72 C. 144 D. 180

Answer: _____

2. What is displayed? [1 mark]

original ← 80
copy ← original
original ← 100
DISPLAY(copy)

A. 80 B. 100 C. 20 D. 180

Answer: _____

3. A file is encoded in fewer bits, and its decoder guarantees recovery of every original byte. Which description is supported? [1 mark]

A. The method is lossy because the representation changed.	B. The method must remove information because it uses fewer bits.
C. The method must make every possible input file smaller.	D. The method is lossless because exact reconstruction is guaranteed.

Answer: _____

4. Using the specified toy count-letter RLE format, which encoding preserves the original sequence AABBBCC? [1 mark]

A. 3A2B2C B. 2A3B2C C. 2A2C3B D. A2B3C2

Answer: _____

5. Which TWO statements about data compression are correct? Select exactly TWO answers. [1 mark]

A. A lossless method always reduces the size of every input.	B. Size reduction depends on the data and the algorithm used.
C. Lossy compression guarantees recovery of every original value.	D. Lossless compression can use fewer bits without losing original information.

Answer: _____

6. A representation changes from 240 bytes to 180 bytes. What percentage of the original size has been saved? [1 mark]

A. 25% B. About 33.3% C. 60% D. 75%

Answer: _____

AP CSP | Lesson 06: Compression

Written responses | Questions 7-9

7. [4 marks]

Use the specified toy RLE for AAAABBBBBBCC.

- (a) Write the encoding.
- (b) State the raw size and encoded size in bytes.
- (c) Explain how the encoding allows exact reconstruction.

8. [4 marks]

- (a) Decode 3C1A2B.
- (b) State the number of reconstructed letters and the encoded size in bytes.
- (c) A student claims 2A2B correctly encodes ABAB because it has two A's and two B's. Explain the error and give a correct encoding for ABAB.

9. [4 marks]

The input is ABCABC. Under the toy format, each raw letter uses one byte and each count-letter pair uses two bytes. No headers are included.

- (a) Write its encoding.
- (b) State its raw and encoded sizes.
- (c) Express the change as a percentage increase from the original.
- (d) Explain why this result does not prove that every compression algorithm will make this input larger.

AP CSP | Lesson 06: Compression

Written responses | Questions 10-12

10. [4 marks]

A raw file is 20 bytes, with no additional header. An encoded version has 10 bytes of payload and needs a 4-byte format header.

- (a) State the total encoded file size.
- (b) State the number of bytes saved compared with the raw file.
- (c) Calculate the percentage saved, showing the denominator.
- (d) Explain why omitting the header would give a misleading comparison.

11. [4 marks]

A toy method stores only the tens digit of each integer measurement from 10 to 99. Its decoder multiplies the stored digit by 10.

- (a) State the stored digit for 31 and for 38.
- (b) State the reconstructed value for each.
- (c) Is this method lossless or lossy? Explain using these two inputs.
- (d) The input 30 is reconstructed correctly. Does this make the method lossless? Explain.

12. [4 marks]

Two encodings of the same image have been measured. Option L is 900 KB and guarantees exact reconstruction. Option P is 160 KB and produces an approximation already approved for preview use.

- (a) An archive requires exact recovery of the original image. Choose an option and justify it.
- (b) A preview system requires a file under 200 KB and accepts the approved approximation. Choose an option and justify it.
